

Supplementary material for Thorp et al. (2026). “Two decades of precision cotton irrigation research data at Maricopa, Arizona” Submitted to Scientific Data.

July 7, 2026

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Table S.1: Data dictionary for items within the “properties” member of GeoJSON objects for 17 experiment-years of precision cotton irrigation research data at Maricopa, Arizona from 2002 to 2023. Variables related to soil, weather, crop, and management data are included, in addition to variables for GeoJSON object labels and unique identifiers. The “Code” column provides the keys to access data within the GeoJSON properties members. The “Type” column provides the data type of the return value for each key. In some cases, the return value is a list of hash tables (or dictionaries), and the “Key : Type” column provides the additional keys and data types of return values for the hash elements in the list. The “Unit or Format” column provides specific information on units of measure or string formatting for the return values. Many variable codes were borrowed from the International Consortium for Agricultural Systems Applications (ICASA) data standards (White et al., 2013), and the “ICASA” column specifies whether or not the variable code was taken from ICASA. The “Class Membership” column provides a list of the data classes in which the variable is found.

Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
ABSNUM	Number of fruiting site abscissions per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
ADAT	Growth stage date, anthesis	str	na	yyyy-mm-dd	Yes	Plot CropCanopy
ADOY	Anthesis date as day of year	int	na	day of year	Yes	Plot CropCanopy
ASCEETO	Standardized grass reference evapotranspiration (Walter et al., 2005)	list of dicts	“date” : str “value” : float	“date” : yyyy-mm-dd “value” : mm	No	Weather
BBFRAC	Fraction of cotton burr weight to mature boll weight	float	na	g g ⁻¹	No	PlantAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
BFFRAC	Fraction of cotton fiber weight to mature boll weight	float	na	g g^{-1}	No	PlantAnalysis
BSFRAC	Fraction of cottonseed weight to mature boll weight	float	na	g g^{-1}	No	PlantAnalysis
BWCAH	Bulk water content of seed cotton at harvest	float	na	%	No	Plot CropHarvest
cc_label	Character string for crop canopy measurement point (unique only within experiment-year)	str	na	na	No	CropCanopy
ccids	Unique IDs of associated CropCanopy features	list of strs	na	BSON ObjectId	No	Plot
ch_label	Character string for crop harvest area (unique only within experiment-year)	str	na	na	No	CropHarvest
CH_NOTES	Information on the application of agrochemicals	list of dicts	“date” : str “value” : str	“date” : yyyy-mm-dd “value” : na	Yes	Plot

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
chids	Unique IDs of associated CropHarvest features	list of strs	na	BSON ObjectId	No	Plot
CHLA	Leaf chlorophyll <i>a</i> content	list of dicts	“date” : str “value” : float	“date” : yyyy-mm-dd “value” : $\mu\text{g cm}^{-2}$	No	CropCanopy
CHTD	Canopy height	list of dicts	“date” : str “value” : float	“date” : yyyy-mm-dd “value” : m	Yes	CropCanopy PlantAnalysis
CMPLC	Complications arising during experiment	str	na	na	Yes	Experiment
CR_SYSTEM	Cropping system descriptive name	str	na	na	Yes	Experiment
CUL_NAME	Cultivar name	str	na	na	No	Plot
CWAD	Dry weight of above ground plant parts	float	na	kg ha^{-1}	Yes	PlantAnalysis
CWID	Canopy width	list of dicts	“date” : str “value” : float	“date” : yyyy-mm-dd “value” : m	Yes	CropCanopy PlantAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
CWPD	Tops dry weight	float	na	g plant ⁻¹	Yes	PlantAnalysis
DBWAH	Dry bulk weight at harvest (fiber + seed + trash) (0% water content)	float	na	kg ha ⁻¹	No	Plot CropHarvest
DFWAH	Dry fiber weight at harvest (0% water content)	float	na	kg ha ⁻¹	No	Plot CropHarvest
DSCWAH	Dry seed cotton weight at harvest (fiber + seed) (0% water content)	float	na	kg ha ⁻¹	No	Plot CropHarvest
DSWAH	Dry seed weight at harvest (0% water content)	float	na	kg ha ⁻¹	No	Plot CropHarvest
EDATE	Date of emergence	str	na	yyyy-mm-dd	Yes	Plot CropCanopy
eid	Unique ID of associated Experiment feature	str	na	BSON ObjectId	No	Plot

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
EM38H	Soil electrical conductivity with EM38 sensor, horizontal orientation	list of dicts	“date” : str “value” : float	“date” : yyyy-mm-dd “value”: mS m ⁻¹	No	Plot Zone CropHarvest
EM38V	Soil electrical conductivity with EM38 sensor, vertical orientation	list of dicts	“date” : str “value” : float	“date” : yyyy-mm-dd “value”: mS m ⁻¹	No	Plot Zone CropHarvest
EX_ADDRESS	Mailing address of researcher	str	na	na	Yes	Experiment
EX_EMAIL	E-mail address of researcher	str	na	na	Yes	Experiment
EXNAME	Name of experiment	str	na	na	Yes	Experiment
EXNDVI	Normalized difference vegetation index (NDVI) from an Exotech handheld radiometer	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: unitless	No	Plot
EXP_DUR	Duration of experiment in years	int	na	years	Yes	Experiment
exp_label	Unique character string for experiment	str	na	na	No	Experiment

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
EXP_NARR	General description of experiment. Can include noteworthy events, findings, etc.	str	na	na	Yes	Experiment
EXP_YEAR	Year associated with experiment or crop rotation	str	na	yyyy	Yes	Experiment
EXPER_TYPE	General type of experiment, e.g., varietal trial	str	na	ICASA code	Yes	Experiment
FACTORS	Experimental factor combinations	str	na	na	Yes	Experiment
FAREA	Field area	float	na	m ²	Yes	Experiment
FBCGR	Fiber color grade from high volume instrument (HVI)	str	na	na	No	Plot CropHarvest
FBCLB	Fiber color +b from HVI	float	na	na	No	Plot CropHarvest
FBCRD	Fiber color reflectance (Rd)	float	na	% reflectance	No	Plot CropHarvest
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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
FBELO	Fiber elongation	float	na	%	No	Plot CropHarvest
FBLTH	Fiber length (i.e., upper-half mean length)	float	na	inches	No	Plot CropHarvest
FBMIC	Fiber micronaire <34 Discount range 35 to 36 Base range 37 to 42 Premium range 43 to 49 Base range >50 Discount range	float	na	none	No	Plot CropHarvest
FBSFI	Short fiber index	float	na	%	No	Plot CropHarvest

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
FBSTR	Fiber strength <23 Weak 24 to 25 Intermediate 26 to 28 Average 29 to 30 Strong >31 Very strong	float	na	g tex^{-1}	No	Plot CropHarvest
FBTAR	HVI trash area as a measure of trash in fiber. Reported as % of the sample surface covered by trash particles as determined by video scanner.	float	na	%	No	Plot CropHarvest
FBTCT	HVI trash count as a measure of trash in fiber. Reported as the number of trash particles visible.	float	na	particles	No	Plot CropHarvest

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
FBUNI	Fiber length uniformity <77 Very low 77 to 79 Low 80 to 82 Average 83 to 85 High >85 Very high	float	na	%	No	Plot CropHarvest
FEAMN	Nitrogen (elemental) in applied fertilizer	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: kg N ha ⁻¹	Yes	Plot
FEAMP	Phosphorus (elemental) in applied fertilizer	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: kg P ha ⁻¹	Yes	Plot
FFRAC	Fiber fraction of bulk harvest after ginning	float	na	g g ⁻¹	No	Plot Zone CropHarvest
FIRST_NAME	First name of researchers involved with the experiment(s) or data	str	na	na	Yes	Experiment

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
FL_LAT	Field latitude, N positive, S negative	float	na	degrees	Yes	Experiment
FL_LONG	Field longitude, E positive, W negative	float	na	degrees	Yes	Experiment
FL_NAME	Field name	str	na	na	Yes	Experiment
FLELE	Elevation of field site	int	na	m	Yes	Experiment
FLRNUM	Number of flowers per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
FRBNUM	Number of fruiting branches per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
FRCOV	Fractional crop cover computed from drone-based orthophotos	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: m ² m ⁻²	No	Plot Zone CropHarvest
FWCAG	Fiber water content after ginning	float	na	%	No	Plot CropHarvest

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
GBNUM	Number of green bolls per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
GNDAT	Ginning date	str	na	yyyy-mm-dd	No	Plot Zone CropHarvest
HADAT	Harvest date	str	na	yyyy-mm-dd	Yes	Plot Zone CropHarvest
HAREA	Harvest sample collection area	float	na	m ²	Yes	CropHarvest
HARM	Harvest method (“Machine” or “Hand”)	str	na	na	Yes	Plot Zone CropHarvest
HTUNT	Heat units, total daily, sine curve method with minimum and maximum air temperature thresholds of 12.8°C and 30°C, respectively (Brown, 2013)	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: °C d ⁻¹	No	Weather
IN_ROLE	Role of participating institutions	str	na	ICASA code	Yes	Experiment

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
IN_TYPE	Type of participating institutions	str	na	ICASA code	Yes	Experiment
INSTITUTION	Names of participating institutions	str	na	na	Yes	Experiment
IROP	Irrigation operation (e.g., furrow, sprinkler)	str	na	ICASA code	Yes	Plot Zone
IRTLST	Land surface temperature from infrared thermometer (soil and canopy composite)	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: °C	No	Plot
IRTLSTCAN	Land surface temperature from infrared thermometer (canopy isolated, no soil)	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: °C	No	Plot
IRVAL	Irrigation amount, depth of water	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: mm	Yes	Plot Zone
LAID_FLD	Leaf area index measured with Li-Cor Plant Canopy Analyzer in the field	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: m ² m ⁻²	No	CropCanopy PlantAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
LAID_LAB	Leaf area index computed from area measurements of leaf samples in the laboratory	float	na	m ² m ⁻²	No	PlantAnalysis
LAST_NAME	Last name of researchers involved with the experiment(s) or data	str	na	na	Yes	Experiment
LF1D	Growth stage date, leaf1 full expansion	str	na	yyyy-mm-dd	No	Plot CropCanopy
LWAD	Leaf dry weight	float	na	kg ha ⁻¹	Yes	PlantAnalysis
LWPD	Leaf dry weight per plant	float	na	g plant ⁻¹	Yes	PlantAnalysis
MAIN_FACTOR	Main factor, factors or interactions of interest in an experiment	str	na	na	Yes	Experiment
MBNUM	Number of mature bolls per plant	float	na	no. plant ⁻¹	No	PlantAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
MBWAD	Mature boll dry weight	float	na	g plant ⁻¹	No	PlantAnalysis
METHODS	General notes on execution of experiment	str	na	na	Yes	Experiment
MGMT_TYPE	Whether managed by researchers, producers	str	na	ICASA code	Yes	Experiment
MID_INITIAL	Middle initial of researchers involved with the experiment(s) or data	str	na	na	Yes	Experiment
MSNODE	Number of mainstem nodes per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
NAWF	Nodes above white flower	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: nodes	No	Plot CropCanopy
OBJECTIVES	Objectives of the study	str	na	na	Yes	Experiment

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
pa_label	Character string for plant analysis sample (unique only within experiment-year)	str	na	na	No	PlantAnalysis
paids	Unique IDs of associated PlantAnalysis features	list of strs	na	BSON ObjectId	No	Plot
PDATE	Planting date	str	na	yyyy-mm-dd	Yes	Plot
PERSON_NOTES	Additional notes on persons involved with the experiment(s)	str	na	na	Yes	Experiment
PFNODE	Number of pre-fruiting mainstem nodes per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
PHTD	Plant height	float	na	m	Yes	PlantAnalysis
pid	Unique ID of associated Plot feature	str	na	BSON ObjectId	No	Zone CropHarvest SoilWaterContent CropCanopy PlantAnalysis SoilChemicalAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
pids	Unique IDs of associated Plot features	list of strs	na	BSON ObjectId	No	Experiment
PLDAY	Day of planting as day of year	int	na	doy	Yes	Plot
PLDOE	Day of year for date of emergence	int	na	doy	Yes	Plot CropCanopy
PLPD	Plant population	float	na	plant m ⁻²	Yes	Plot CropCanopy
plt_label	Character string for plot (unique only within experiment-year)	str	na	na	No	Plot
PLTA	Plot area	float	na	m ²	Yes	Plot
PLTCNT	Number of plants in sample	int	na	plants	No	PlantAnalysis
PLYR	Planting year	int	na	yyyy	Yes	Plot
PLYRE	Year for day of emergence	int	na	yyyy	Yes	Plot CropCanopy
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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
PSDATE	Plant sampling date	str	na	yyyy-mm-dd	Yes	PlantAnalysis
PWAD	Pod (boll) dry weight	float	na	kg ha ⁻¹	Yes	PlantAnalysis
QLDAT	Fiber quality measurement date	str	na	yyyy-mm-dd	No	Plot CropCanopy
RAIN	Precipitation, total daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: mm	Yes	Weather
REFET	Reference evapotranspiration, original AZMET	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: mm	No	Weather
REFHT	Reference height for weather measurement	float	na	m	Yes	Weather
REP_NO	Number of blocks or replicates	int	na	replicates	Yes	Experiment
RHAVD	Relative humidity, average daily value	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: %	Yes	Weather

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
RHMXD	Relative humidity, maximum daily value	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: %	Yes	Weather
RHUMD	Relative humidity at maximum daily temperature (or minimum daily relative humidity)	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: %	Yes	Weather
RSLST	Land surface temperature from airborne remote sensing	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: °C	No	Plot Zone CropHarvest
RSNDVI	Normalized difference vegetation index from airborne remote sensing	list of dicts	“date”: str “value”: float	“date” : yyyy-mm-dd “value”: unitless	No	Plot Zone CropHarvest
sca_label	Character string for the soil sample (unique only within experiment-year)	str	na	na	No	SoilChemicalAnalysis
scaids	Unique IDs of associated SoilChemicalAnalysis features	list of strs	na	BSON ObjectId	No	Plot
SDPB	Number of seeds per boll	float	na	seed boll ⁻¹	No	PlantAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SFRAC	Seed fraction of bulk harvest after ginning	float	na	g g^{-1}	No	Plot Zone CropHarvest
SITE_NAME	Name of site for experiment	str	na	na	Yes	Experiment
SITE_TYPE	Type of site, e.g., station, on-farm	str	na	ICASA code	Yes	Experiment
SLCA	Soil calcium concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: int	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLCEC	Cation exchange capacity	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: cmol kg^{-1}	Yes	SoilChemicalAnalysis
SLCLY_GB	Soil texture, clay percentage, Gee and Bauder (1986) hydrometer and calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SLCLY_GBM	Soil texture, clay percentage, Gee and Bauder (1986) hydrometer method and Maricopa in-house calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis
SLCLY_M	Soil texture, clay percentage, Maricopa in-house hydrometer and calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis
SLCU	Soil copper concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	No	SoilChemicalAnalysis
SLEC	Soil electrical conductivity	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: mS cm ⁻¹	Yes	SoilChemicalAnalysis
SLFC1	Soil water content at field capacity (1/3 bars suction)	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: cm ³ cm ⁻³	Yes	Plot Zone CropHarvest SoilPhysicalAnalysis
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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SLFE	Soil iron concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLKE	Soil potassium concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: int	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLMG	Soil magnesium concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: int	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLMN	Soil manganese concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLNA	Soil sodium concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: int	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SLOM	Soil organic matter	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	Yes	SoilChemicalAnalysis
SLPHB	Soil pH in buffer	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: unitless	Yes	SoilChemicalAnalysis
SLPHW	Soil pH in water	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: unitless	Yes	SoilChemicalAnalysis
SLPX	Soil phosphorus, extractable	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLSLT_GB	Soil texture, silt percentage, Gee and Bauder (1986) hydrometer and calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SLSLT_GBM	Soil texture, silt percentage, Gee and Bauder (1986) hydrometer method and Maricopa in-house calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis
SLSLT_M	Soil texture, silt percentage, Maricopa in-house hydrometer and calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis
SLSND_GB	Soil texture, sand percentage, Gee and Bauder (1986) hydrometer and calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis
SLSND_GBM	Soil texture, sand percentage, Gee and Bauder (1986) hydrometer method and Maricopa in-house calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SLSND_M	Soil texture, sand percentage, Maricopa in-house hydrometer and calculation method	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: %	No	Plot Zone CropHarvest SoilPhysicalAnalysis
SLSU	Soil sulfur concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	Yes	SoilChemicalAnalysis
SLTX	Soil texture	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: str	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: none	Yes	SoilPhysicalAnalysis
SLWP	Soil water content at wilting point (15 bars suction)	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: cm ³ cm ⁻³	Yes	Plot Zone CropHarvest SoilPhysicalAnalysis
SLWP2	Soil water content at 5 bars suction	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: cm ³ cm ⁻³	No	Plot Zone CropHarvest SoilPhysicalAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SLZN	Soil zinc concentration	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: ppm	No	SoilChemicalAnalysis
SNH4	Soil ammonium	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: g Mg ⁻¹	Yes	SoilChemicalAnalysis
SNO3	Soil nitrate	list of dicts	“depth”: int “mindepth”: int “maxdepth”: int “value”: float	“depth”: cm “mindepth”: cm “maxdepth”: cm “value”: g Mg ⁻¹	Yes	SoilChemicalAnalysis
SOIL_DATE	Soil sampling date in field	str	na	yyyy-mm-dd	Yes	SoilChemicalAnalysis SoilPhysicalAnalysis
spa_label	Character string for the soil sample (unique only within experiment-year)	str	na	na	No	SoilPhysicalAnalysis
SPAD	SPAD number from SPAD meter	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: SPAD units	No	CropCanopy

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
spaid5	Unique IDs of associated SoilPhysicalAnalysis features	list of str5	na	BSON ObjectId	No	Plot
SQRNUM	Number of squares per plant	float	na	no. plant ⁻¹	No	PlantAnalysis
SRAD	Solar radiation, total daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: MJ m ⁻²	Yes	Weather
STAV1	Soil temperature, average daily at 4 inch soil depth (2 inch prior to 1999)	list of dicts	“date”: str “depth”: float “value”: float	“date”: yyyy-mm-dd “depth”: cm “value”: °C	No	Weather
STAV2	Soil temperature, average daily at 20 inch soil depth (4 inch prior to 1999)	list of dicts	“date”: str “depth”: float “value”: float	“date”: yyyy-mm-dd “depth”: cm “value”: °C	No	Weather
STDD	Stem diameter	float	na	mm	No	PlantAnalysis
STMN1	Soil temperature, minimum daily at 4 inch soil depth (2 inch prior to 1999)	list of dicts	“date”: str “depth”: float “value”: float	“date”: yyyy-mm-dd “depth”: cm “value”: °C	No	Weather

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
STMN2	Soil temperature, minimum daily at 20 inch soil depth (4 inch prior to 1999)	list of dicts	“date”: str “depth”: float “value”: float	“date”: yyyy-mm-dd “depth”: cm “value”: °C	No	Weather
STMX1	Soil temperature, maximum daily at 4 inch soil depth (2 inch prior to 1999)	list of dicts	“date”: str “depth”: float “value”: float	“date”: yyyy-mm-dd “depth”: cm “value”: °C	No	Weather
STMX2	Soil temperature, maximum daily at 20 inch soil depth (4 inch prior to 1999)	list of dicts	“date”: str “depth”: float “value”: float	“date”: yyyy-mm-dd “depth”: cm “value”: °C	No	Weather
SUITE_NAME	Brief name of suite of experiments	str	na	na	Yes	Experiment
SUITE_OBJ	Objectives of the study	str	na	na	Yes	Experiment
SWAD	Stem dry weight	float	na	kg ha ⁻¹	Yes	PlantAnalysis
swc_label	Character string for access tube (unique only within experiment-year)	str	na	na	No	SoilWaterContent

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
SWCAG	Seed water content after ginning	float	na	%	No	Plot CropHarvest
swcids	Unique IDs of associated SoilWaterContent features	list of strs	na	BSON ObjectId	No	Plot
SWLD	Soil water content	list of dicts	“date”: str “type”: str “depth”: int “mindepth”: int “maxdepth”: int “value”: float	“date” : yyyy-mm-dd “type” na “depth”: cm “mindepth”: cm “maxdepth”: cm “value”: cm ³ cm ⁻³	Yes	SoilWaterContent
SWPD	Stem dry weight per plant	float	na	g plant ⁻¹	Yes	PlantAnalysis
TAVD	Temperature of air, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: °C	Yes	Weather
TDEW	Dew point temperature, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: °C	Yes	Weather
TFRAC	Trash fraction of bulk harvest after ginning	float	na	g g ⁻¹	No	Plot Zone CropHarvest

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
TLNOTES	Additional, detailed notes on tillage practices	list of dicts	“date”: str “value”: str	“date”: yyyy-mm-dd “value”: na	Yes	Plot
TMAX	Temperature of air, daily maximum	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: °C	Yes	Weather
TMIN	Temperature of air, daily minimum	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: °C	Yes	Weather
TRT_COUNT	Number of treatments	int	na	treatments	Yes	Experiment
trt_info	Treatment descriptions and plot assignments	list of dicts	“trt_label”: str “description”: str “pids”: list of strs	“trt_label”: na “description”: na “pids”: BSON ObjectId	No	Experiment
trt_label	Character string for treatment (unique only within experiment-year)	str	na	na	No	Plot
VBNODE	Number of nodes on vegetative branches	float	na	nodes	No	PlantAnalysis

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
VBNUM	Number of vegetative branches	float	na	branches	No	PlantAnalysis
VPDAVD	Vapor pressure deficit, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: kPa	No	Weather
VPRSD	Vapor pressure, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: kPa	Yes	Weather
w_label	Character string for weather station	str	na	na	No	Weather
WBWAH	Wet bulk weight at harvest (fiber + seed + trash)	float	na	kg ha ⁻¹	No	Plot Zone CropHarvest
WFWAH	Wet fiber weight at harvest	float	na	kg ha ⁻¹	No	Plot Zone CropHarvest
WNDD	Wind vector, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: degrees	No	Weather
WNDDSD	Wind vector, standard deviation daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: degrees	No	Weather
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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
WNDHT	Reference height for wind speed measurement	float	na	m	Yes	Weather
WNDMX	Wind speed, maximum daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: m s ⁻¹	No	Weather
WNDRN	Wind run, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: km d ⁻¹	No	Weather
WNDSP	Wind speed, average daily	list of dicts	“date”: str “value”: float	“date”: yyyy-mm-dd “value”: m s ⁻¹	No	Weather
WSCWAH	Wet seed cotton weight at harvest (fiber + seed)	float	na	kg ha ⁻¹	No	Plot Zone CropHarvest
WSWAH	Wet seed weight at harvest	float	na	kg ha ⁻¹	No	Plot Zone CropHarvest
zids	Unique IDs of associated Zone features	list of strs	na	BSON ObjectId	No	Plot
ZON_AREA	Area of management zone	float	na	m ²	No	Zone

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Code	Description	Type	Key : Type	Unit or Format	ICASA	Class Membership
zon_label	Character string for zone (unique only within experiment-year)	str	na	na	No	Zone

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